

# WILLIAM WU

William.wu1239@gmail.com | 408-832-8628 | [Portfolio](#)

Computer Science graduate building full-stack AI, data, and web applications with Python, FastAPI, JavaScript, TensorFlow, and modern frontend tooling.

## TECHNICAL SKILLS

---

**Languages:** Python, C++, Java, JavaScript, TypeScript, HTML/CSS

**Frameworks & Libraries:** FastAPI, TensorFlow, Keras, scikit-learn, pandas, NumPy, Phaser, Next.js, Prisma, Tailwind CSS

**Databases & Tools:** SQLite, Git, Playwright, Figma, Wireframing

**AI/ML:** Diffusion models, computer vision, machine learning workflows, data preprocessing, model evaluation

**Strengths:** Full-stack development, data analysis, UI/UX thinking, problem solving

## PROJECTS

---

### Real Estate Visualizer - AI Image Editing Web Application | [GitHub](#)

*Python, FastAPI, JavaScript, HTML/CSS, Diffusion Models, Computer Vision*

- Built a full-stack AI image editing web application for virtual home staging using a FastAPI backend and browser-based frontend.
- Integrated diffusion-based generative image models through Hugging Face to support image generation and targeted editing workflows.
- Implemented mask-based image manipulation features that let users add, remove, and modify objects within real estate photos.
- Developed image-processing pipelines for upload handling, preprocessing, model inference, and post-processing while preserving original room layouts.

### REI Opportunity Finder - Real Estate Investment Analysis Tool | [GitHub](#)

*TypeScript, Next.js, Prisma, SQLite, Playwright, Tailwind CSS*

- Built a full-stack real estate investment tool that scrapes active listings from Redfin and Zillow using headless browser automation with stealth detection bypass.
- Implemented a property scoring algorithm (0-100) based on days on market, price reductions, motivation keywords, and comparable sales data to surface high-potential deals.
- Estimated property values using weighted-median analysis of recently sold comparable properties with Haversine distance calculations for proximity filtering.
- Added offer-range suggestions, customized seller outreach email generation, and a pipeline tracker to manage deals from discovery through contract negotiation.

### Multi-Label Weather Condition Predictor | [Colab](#)

*Python, TensorFlow, pandas, scikit-learn*

- Built a neural network to classify daily weather conditions from hourly temperature, humidity, wind, and pressure data.
- Cleaned and merged real-world datasets with pandas, normalized features with StandardScaler, and encoded labels with MultiLabelBinarizer.
- Trained a Keras model with dropout layers and evaluated results using classification metrics, Hamming loss, and daily prediction outputs.

### Wreck-It-Ralph Game Clone | [Live Demo](#)

*JavaScript, HTML5, Phaser*

- Developed a browser-based 2D game inspired by Wreck-It Ralph, featuring Play, Menu, and Game Over scenes.
- Implemented character movement, collision logic, reusable prefabs, and game state transitions using the Phaser framework.
- Deployed a complete HTML5 web game with bundled assets for browser-based play.

## EDUCATION

---

### University of California, Santa Cruz - Bachelor of Science in Computer Science

## PROFESSIONAL EXPERIENCE

---

### Real Estate Agent | 2020 - Present

*Client Relations, Negotiation, Customer Service*

- Worked directly with clients to understand needs, evaluate options, address concerns, and support property-related decisions.
- Built communication, negotiation, and stakeholder-management skills through ongoing client interaction and issue resolution.
- Applied structured problem solving in fast-paced situations requiring responsiveness, organization, and attention to detail.
- Applied real estate domain knowledge to identify workflow problems later addressed through software tools for property analysis and visualization.